intertidal zone of British Columbia. The crows put lots of effort into digging clams out of the mud, opening the shells, and feeding on the animal inside. The ecologists noticed that smaller clams went unopened and concluded that the crows had to make an energy trade-off between handling time and edible food. The time and energy needed to open up small clams was better spent digging for another, larger clam. A similar study with oystercatchers and mussels found that the largest mussels were left—they had thicker, barnacle-clad shells, so opening them was more difficult. The oystercatchers benefited more by looking for thin-shelled mussels, despite their smaller size.

Animals also have to make choices about where and when to feed. The longer a starling spends in one patch of suitable grassland, for example, the harder it will become to find prey, so it has to decide when to abandon that patch and move to another—an example of what is known as the “marginal value theorem.” Foraging animals also need to consider a range of other factors such as the presence of predators, the number of animals competing for the same food, and the impact of human activity.